

Lab 3 – Architecture Selection & Justification

System: Self-Service Coffee Kiosk

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Architecture Selection:

We chose **Layered Architecture** for the Self-Service Coffee Kiosk System.

1. Architectural Choice

Layered Architecture separates the system into Presentation, Business, and Data layers. The Presentation layer handles the touch-screen interaction, the Business layer manages ordering and payment processing, and the Data layer stores menu and pricing information.

2. Reasons for Selection

Reason 1 – Separation of responsibilities:

The architecture separates customer interaction from business processing and data storage. This makes components such as the User Interface, Order Manager, and Database easier to organize and maintain.

Reason 2 – Hardware and service integration:

The architecture allows the business layer to interact with services such as the Payment Service and Receipt Printer without placing all functionality in the user interface.

3. Security Advantage

Separating the presentation, business, and data responsibilities helps restrict direct access to stored menu and pricing data. The business layer can control which operations are allowed before data access occurs.

4. Performance Benefit

Separating responsibilities allows frequently used operations such as menu retrieval and order processing to be handled by their respective components, reducing unnecessary processing in the touch-screen interface